AWS ARCHITECT ASSOCIATE • LAB VERIFICATION REPORT

AWS VPC Peering & Private Connectivity Report

Implementation, Routing Table Analysis, Bastion ProxyJump Administration & Controlled Validation

STUDENT NAME	BATCH NAME	REGION & AZ	COMPLETION DATE
Abhishek Bohra	**Aws Internship Batch 1**	**ap-southeast-2 (2a)**	**30/10/2026**

1. Executive Summary & Architecture Overview

Administrative SSH management is handled through a hardened public bastion host in VPC A using OpenSSH `ProxyJump`, preventing key exposure on intermediate jump boxes. Inter-workload data exchanges between `App-A` and `App-B` are routed privately across an AWS VPC Peering Connection (`pcx-0db8756df5679b159`).

1.1 Target Architecture & Subnet Allocations

VPC	CIDR Block	Subnet Name	Subnet ID	Subnet CIDR	Associated Workload	Availability Zone
VPC A (Requester)	10.10.0.0/16	ewb-a-public	subnet-029dd74554eb2ba10	10.10.1.0/24	Bastion Host (10.10.1.10)	ap-southeast-2a
VPC A (Requester)	10.10.0.0/16	ewb-a-private	subnet-0c35669863647506b	10.10.2.0/24	App-A Server (10.10.2.10)	ap-southeast-2a
VPC B (Accepter)	10.20.0.0/16	ewb-b-private	subnet-01ab8336092d99d46	10.20.1.0/24	App-B Server (10.20.1.10)	ap-southeast-2a

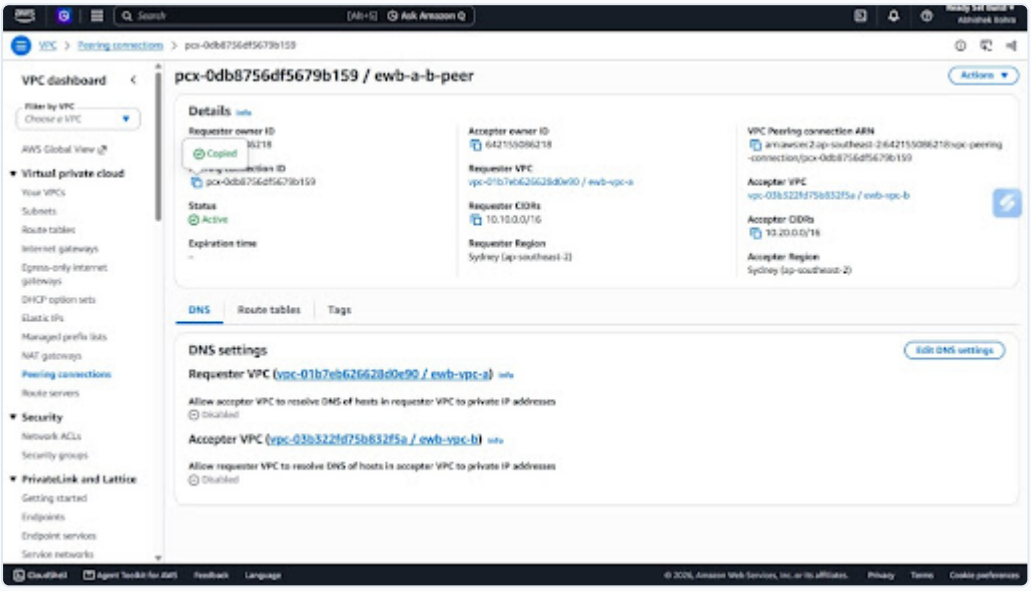
> **Non-Overlapping CIDR Validation:** The CIDR blocks `10.10.0.0/16` and `10.20.0.0/16` are mathematically disjoint. AWS VPC Peering strictly prohibits overlapping address spaces. Subnet ranges alone cannot resolve an overlap if the parent VPC CIDRs collide.

2. AWS Resource Inventory

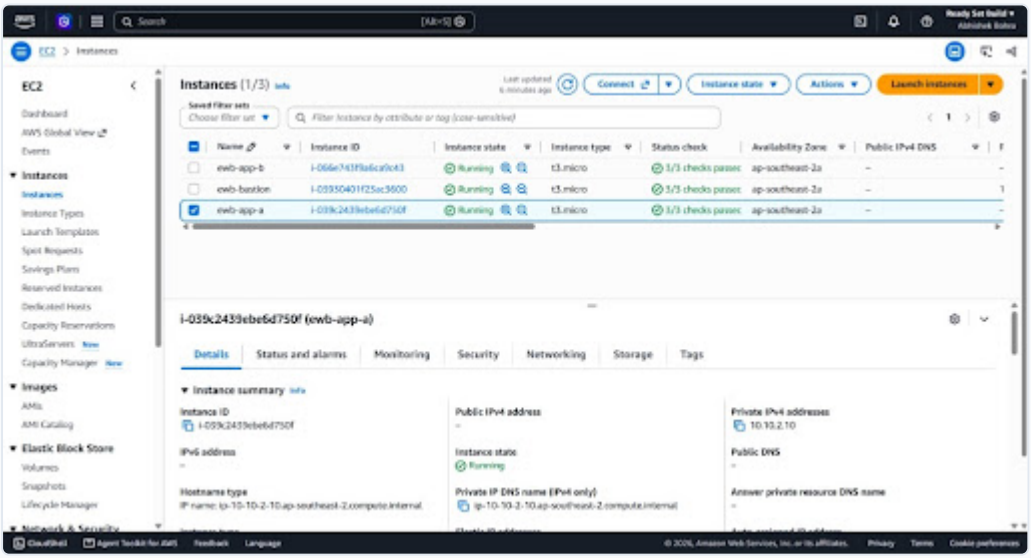
All resources were tagged with `Project=EWB-VPC-Peering` and `Student=Abhishek Bohra`.

Resource Name Tag	AWS Resource ID	Private IPv4	Public IPv4	Status
ewb-vpc-a	vpc-01b7eb626628d0e90	10.10.0.0/16	N/A	Available
ewb-vpc-b	vpc-03b322fd75b832f5a	10.20.0.0/16	N/A	Available
ewb-igw-a	igw-081a3a5d780167876	Attached to VPC A	N/A	Attached
ewb-a-b-peer	pcx-0db8756df5679b159	10.10.0.0/16 ↔ 10.20.0.0/16	N/A	Active

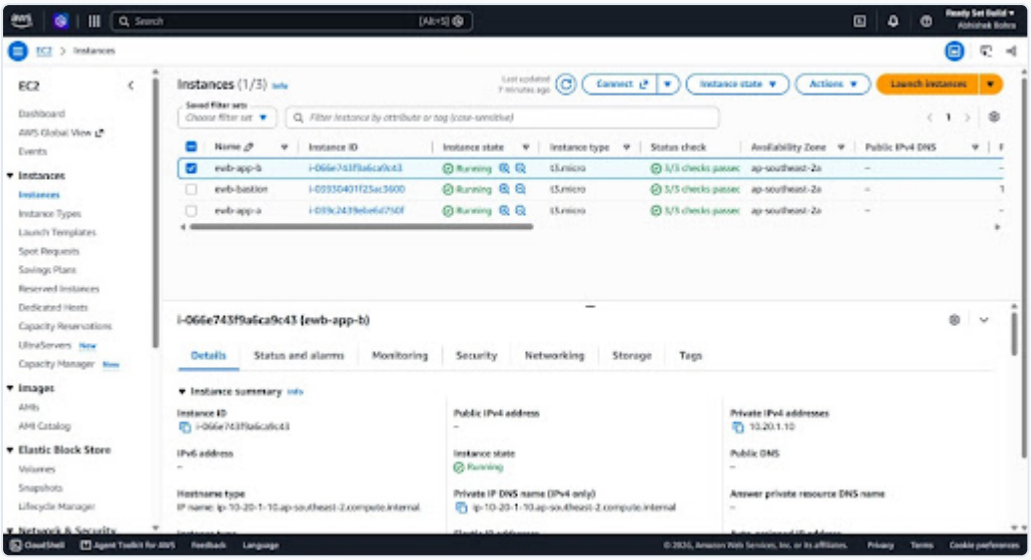
Resource Name Tag	AWS Resource ID	Private IPv4	Public IPv4	Status
ewb-bastion	i-03930401f23ac3600	10.10.1.10	15.135.243.160	Running / 3/3 Passed
ewb-app-a	i-039c2439ebe6d750f	10.10.2.10	None (Isolated)	Running / 3/3 Passed
ewb-app-b	i-066e743f9a6ca9c43	10.20.1.10	None (Isolated)	Running / 3/3 Passed



Evidence Artifact 01: Peering Connection ewb-a-b-peer (pcx-0db8756df5679b159) in Active status between vpc-01b7eb626628d0e90 and vpc-03b322fd75b832f5a.



Evidence Artifact 02: EC2 Instance ewb-app-a (i-039c2439ebe6d750f) running with private IP 10.10.2.10 and no public IPv4 address.



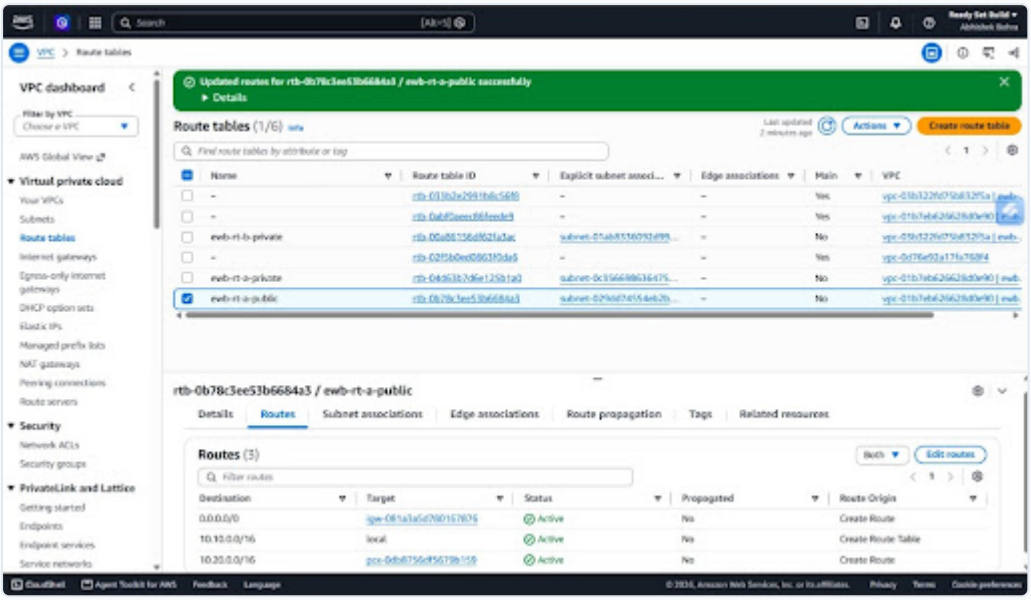
Evidence Artifact 03: EC2 Instance ewb-app-b (i-066e743f9a6ca9c43) running with private IP 10.20.1.10 and no public IPv4 address.

3. Route Tables & Subnet Association Architecture

A VPC peering connection functions solely as a passive network conduit. Active routing requires explicit entries in the respective route tables. In symmetrical routing, each VPC requires a return route back to the sender's CIDR block.

3.1 Public Route Table: ewb-rt-a-public (ID: rtb-0b78c3ee53b6684a3 | Subnet: ewb-a-public)

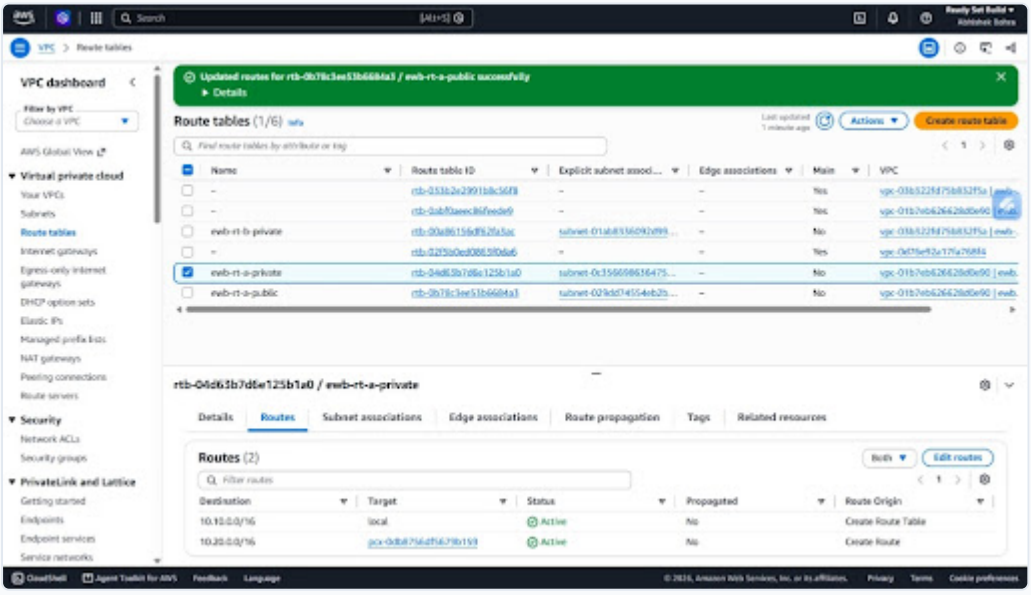
Destination CIDR	Target	Status	Architectural Purpose
10.10.0.0/16	local	Active	Intra-VPC A communication (Bastion to App-A).
0.0.0.0/0	igw-081a3a5d780167876	Active	Inbound administrator SSH & outbound internet access.
10.20.0.0/16	pcx-0db8756df5679b159	Active	Enables Bastion to establish SSH ProxyJump connection to App-B.



Evidence Artifact 04: Route table ewb-rt-a-public showing routes to local, IGW, and the peering target pcx-0db8756df5679b159.

3.2 Private Route Table: ewb-rt-a-private (ID: rtb-04d63b7d6e125b1a0 | Subnet: ewb-a-private)

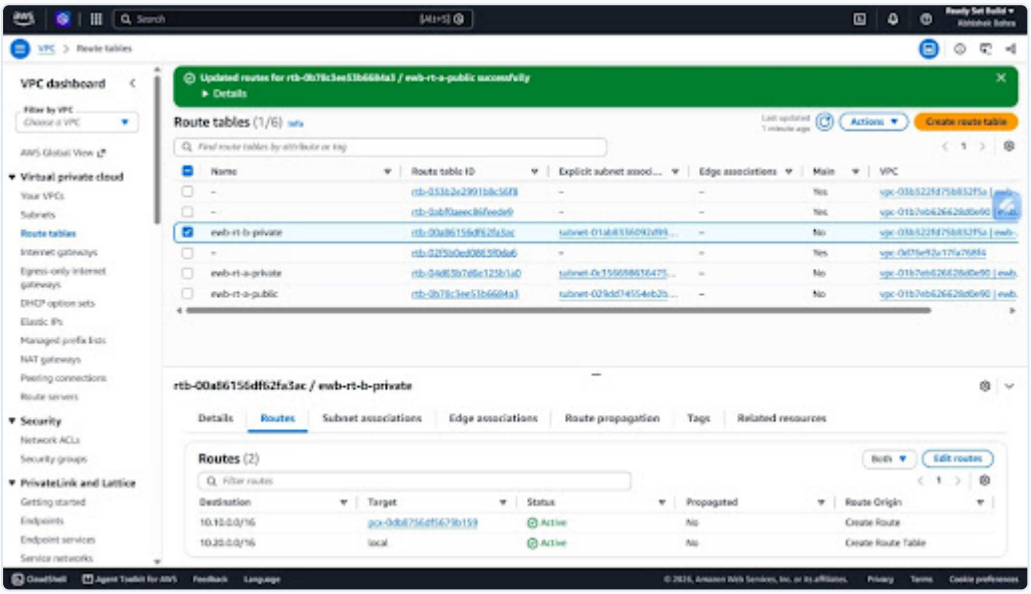
Destination CIDR	Target	Status	Architectural Purpose
10.10.0.0/16	local	Active	Local VPC communication. No internet route configured.
10.20.0.0/16	pcx-0db8756df5679b159	Active	Routes forward private traffic from App-A to VPC B workloads.



Evidence Artifact 05: Route table ewb-rt-a-private (rtb-04d63b7d6e125b1a0) with active route to 10.20.0.0/16 targeting pcx-0db8756df5679b159.

3.3 Private Route Table: ewb-rt-b-private (ID: rtb-00a86156df62fa3ac | Subnet: ewb-b-private)

Destination CIDR	Target	Status	Architectural Purpose
10.20.0.0/16	local	Active	Local VPC B communication. No internet gateway attached to VPC B.
10.10.0.0/16	pcx-0db8756df5679b159	Active	**Return Path:** Routes responses and SSH return packets back to VPC A.



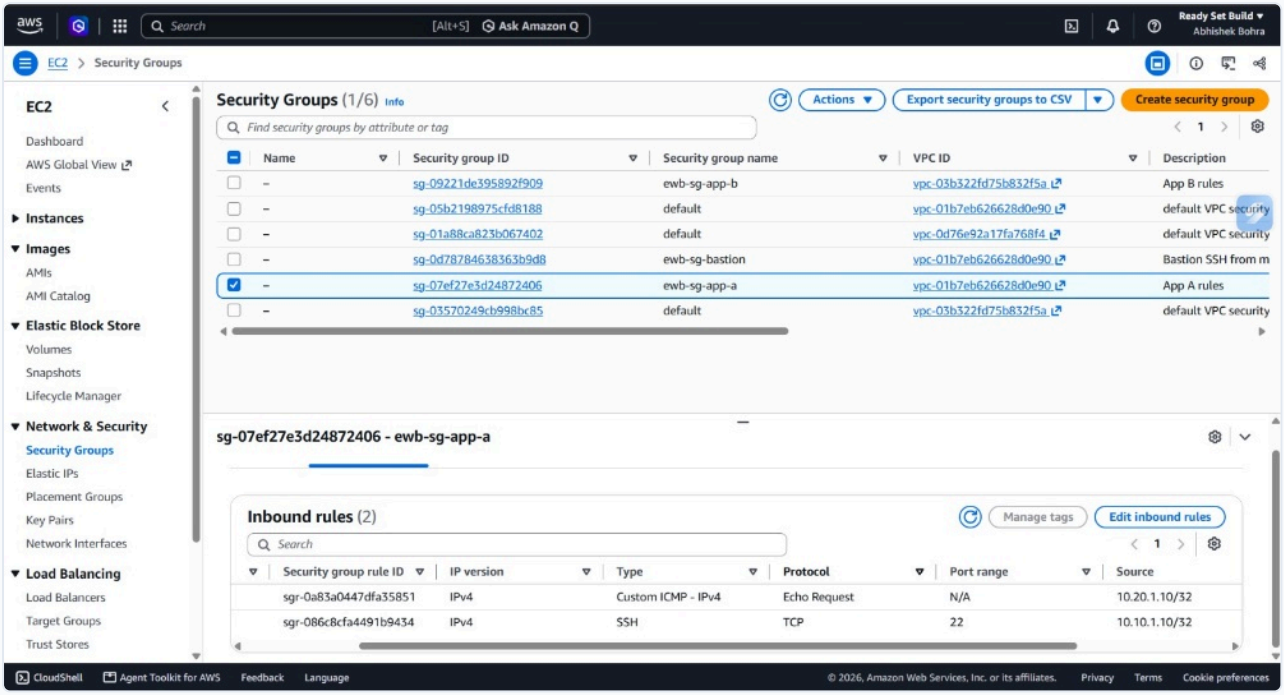
Evidence Artifact 06: Route table ewb-rt-b-private (rtb-00a86156df62fa3ac) with explicit return route to 10.10.0.0/16 targeting pcx-0db8756df5679b159.

4. Security Group Configurations

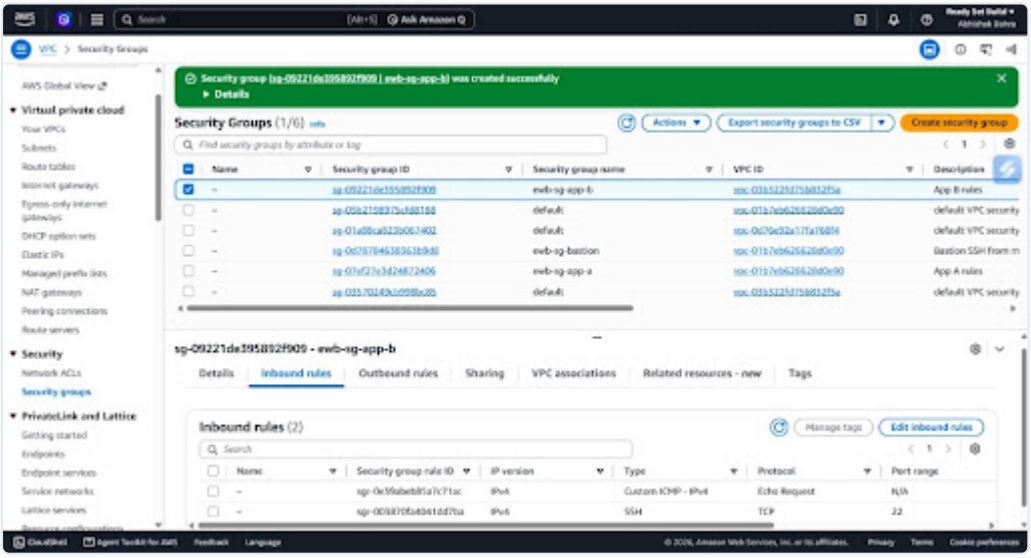
In accordance with least-privilege principles, all security groups scope inbound access to specific `/32` host IPs. Outbound rules retain default `0.0.0.0/0` allow-all since AWS Security Groups are stateful.

Security Group	Security Group ID	Type / Protocol	Port Range	Source Scope	Security Rationale
ewb-sg-bastion	sg-0d78784638363b9d8	SSH (TCP)	22	Admin Public IP /32	Restricts bastion access to authorized administrator workstation.
ewb-sg-app-a	sg-07ef27e3d24872406	SSH (TCP)	22	10.10.1.10/32	Permits administration only via the private IP of the Bastion.
ewb-sg-app-a	sg-07ef27e3d24872406	Custom ICMP - IPv4	Echo Request (8/0)	10.20.1.10/32	Authorizes inbound diagnostic pings solely from App-B.
ewb-sg-app-b	sg-09221de395892f909	SSH (TCP)	22	10.10.1.10/32	Permits ProxyJump connections originating from the Bastion.

Security Group	Security Group ID	Type / Protocol	Port Range	Source Scope	Security Rationale
ewb-sg-app-b	sg-09221de395892f909	Custom ICMP - IPv4	Echo Request (8/0)	10.10.2.10/32	Authorizes inbound diagnostic pings solely from App-A.



Evidence Artifact 07: Security Group ewb-sg-app-a (sg-07ef27e3d24872406) showing SSH from 10.10.1.10/32 and ICMP from 10.20.1.10/32.



Evidence Artifact 08: Security Group ewb-sg-app-b (sg-09221de395892f909) showing SSH from 10.10.1.10/32 and ICMP from 10.10.2.10/32.

5. Administration via OpenSSH ProxyJump

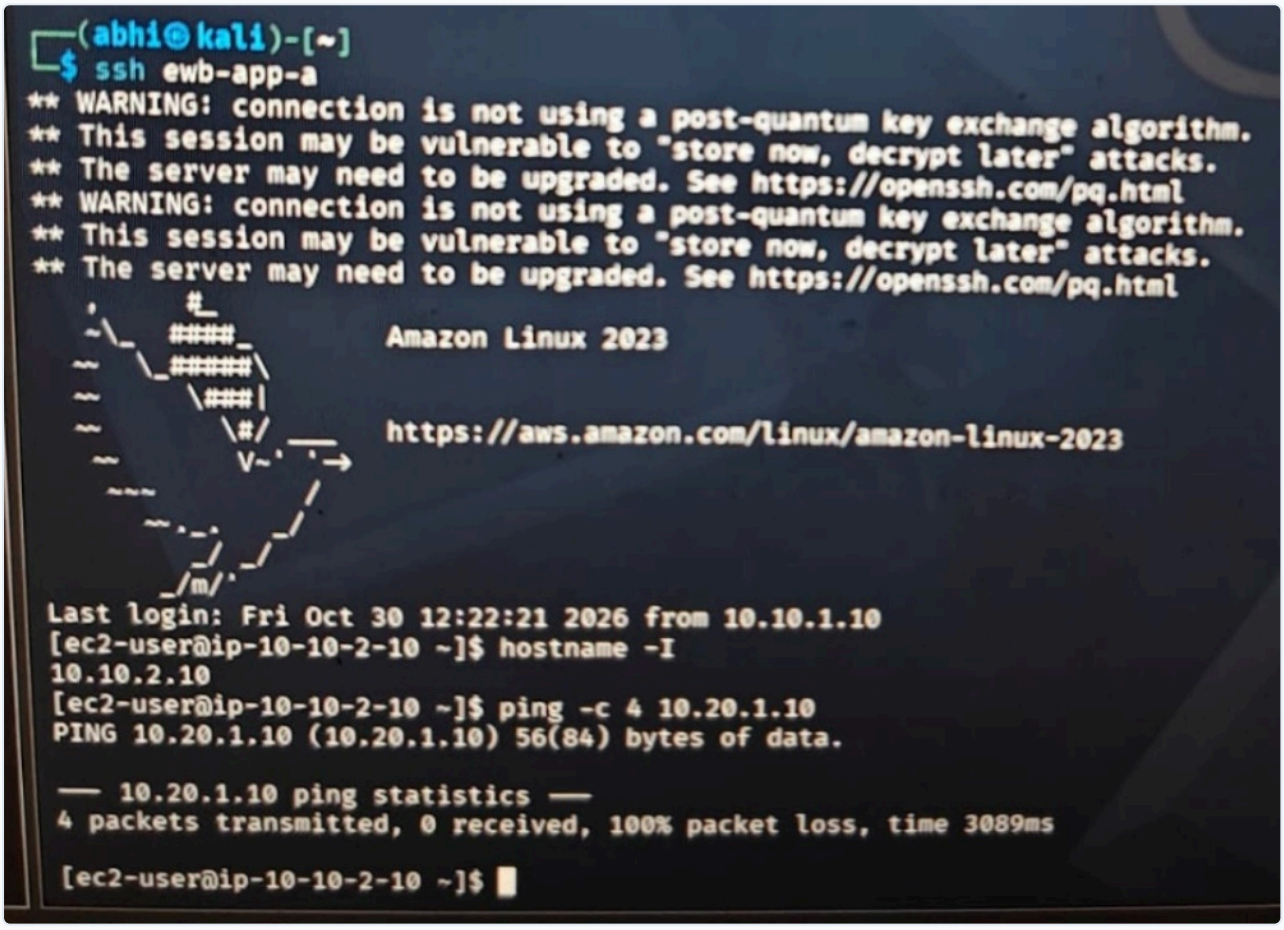
To access the completely private instances without exposing private keys on intermediate servers, OpenSSH client-side `ProxyJump` was configured on the local administrator workstation.

```
# ~/.ssh/config configuration Host ewb-bastion HostName 15.135.243.160 User ec2-user IdentityFile ~/.ssh/ewb-lab.pem IdentitiesOnly yes Host ewb-app-a HostName 10.10.2.10 User ec2-user IdentityFile ~/.ssh/ewb-lab.pem IdentitiesOnly yes ProxyJump ewb-bastion Host ewb-app-b HostName 10.20.1.10 User ec2-user IdentityFile ~/.ssh/ewb-lab.pem IdentitiesOnly yes ProxyJump ewb-bastion
```

6. Positive, Negative & Controlled Failure Verification

Test 01: Baseline Negative Test (Before Route & Peering Activation)

Executed from App-A attempting to reach App-B (`10.20.1.10`). Confirms complete network isolation prior to peering and route table entries.



Evidence Artifact 09: Baseline test on App-A showing 100% packet loss to 10.20.1.10 prior to peering configuration.

```
┌──(abhi㉿kali)-[~] └─$ ssh ewb-app-a Last login: Fri Oct 30 12:22:21 2026 from 10.10.1.10 [ec2-user@ip-10-10-2-10 ~]$ hostname -I 10.10.2.10 [ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10 PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data. --- 10.20.1.10 ping statistics --- 4 packets transmitted, 0 received, 100% packet loss, time 3089ms
```

Test 02: Forward Path Verification (App-A → App-B)

Executed from App-A after peering activation, route table population, and security group verification.

```
┌──(abhi㉿kali)-[~]
└─$ ssh ewb-app-a
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to 'store now, decrypt later' attacks.
** The server may need to be upgraded. See https://openssh.com/pg.html
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to 'store now, decrypt later' attacks.
** The server may need to be upgraded. See https://openssh.com/pg.html
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V-/.'-\
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Fri Oct 30 12:22:21 2026 from 10.10.1.10
[ec2-user@ip-10-10-2-10 ~]$ hostname -I
10.10.2.10
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
64 bytes from 10.20.1.10: icmp_seq=1 ttl=63 time=3091 ms
64 bytes from 10.20.1.10: icmp_seq=2 ttl=63 time=3088 ms
64 bytes from 10.20.1.10: icmp_seq=3 ttl=63 time=3087 ms
64 bytes from 10.20.1.10: icmp_seq=4 ttl=63 time=3089 ms

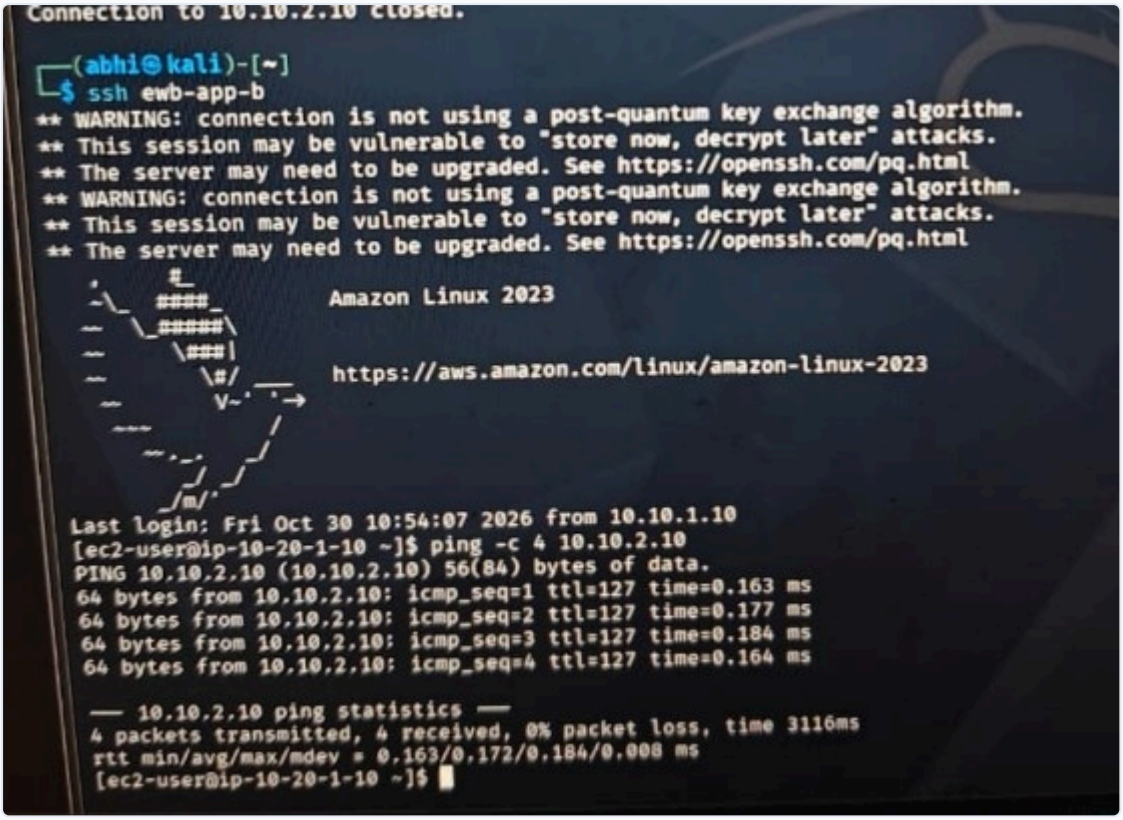
--- 10.20.1.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3089ms
[ec2-user@ip-10-10-2-10 ~]$
```

Evidence Artifact 10: Successful forward ping from App-A (10.10.2.10) to App-B (10.20.1.10) confirming 0% packet loss.


```
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10 PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data. 64 bytes from 10.20.1.10: icmp_seq=1 ttl=63 time=3091 ms 64 bytes from 10.20.1.10: icmp_seq=2 ttl=63 time=3088 ms 64 bytes from 10.20.1.10: icmp_seq=3 ttl=63 time=3087 ms 64 bytes from 10.20.1.10: icmp_seq=4 ttl=63 time=3089 ms --- 10.20.1.10 ping statistics --- 4 packets transmitted, 4 received, 0% packet loss, time 3089ms
```


Test 03 & 04: Reverse Path (App-B → App-A) & Transparent SSH ProxyJump

Executed from Kali workstation jumping directly into App-B (`10.20.1.10`) via Bastion (`10.10.1.10`), then initiating an ICMP Echo Request targeting App-A (`10.10.2.10`).



Evidence Artifact 11: Reverse ping from App-B (10.20.1.10) to App-A (10.10.2.10) proving symmetrical return connectivity and OpenSSH ProxyJump reachability.

```
┌──(abhi㉿kali)-[~] └─$ ssh ewb-app-b Last login: Fri Oct 30 10:54:07 2026 from 10.10.1.10 [ec2-user@ip-10-20-1-10 ~]$ ping -c 4 10.10.2.10 PING 10.10.2.10 (10.10.2.10) 56(84) bytes of data. 64 bytes from 10.10.2.10: icmp_seq=1 ttl=127 time=0.163 ms 64 bytes from 10.10.2.10: icmp_seq=2 ttl=127 time=0.177 ms 64 bytes from 10.10.2.10: icmp_seq=3 ttl=127 time=0.184 ms 64 bytes from 10.10.2.10: icmp_seq=4 ttl=127 time=0.164 ms --- 10.10.2.10 ping statistics --- 4 packets transmitted, 4 received, 0% packet loss, time 3116ms rtt min/avg/max/mdev = 0.163/0.172/0.184/0.008 ms
```

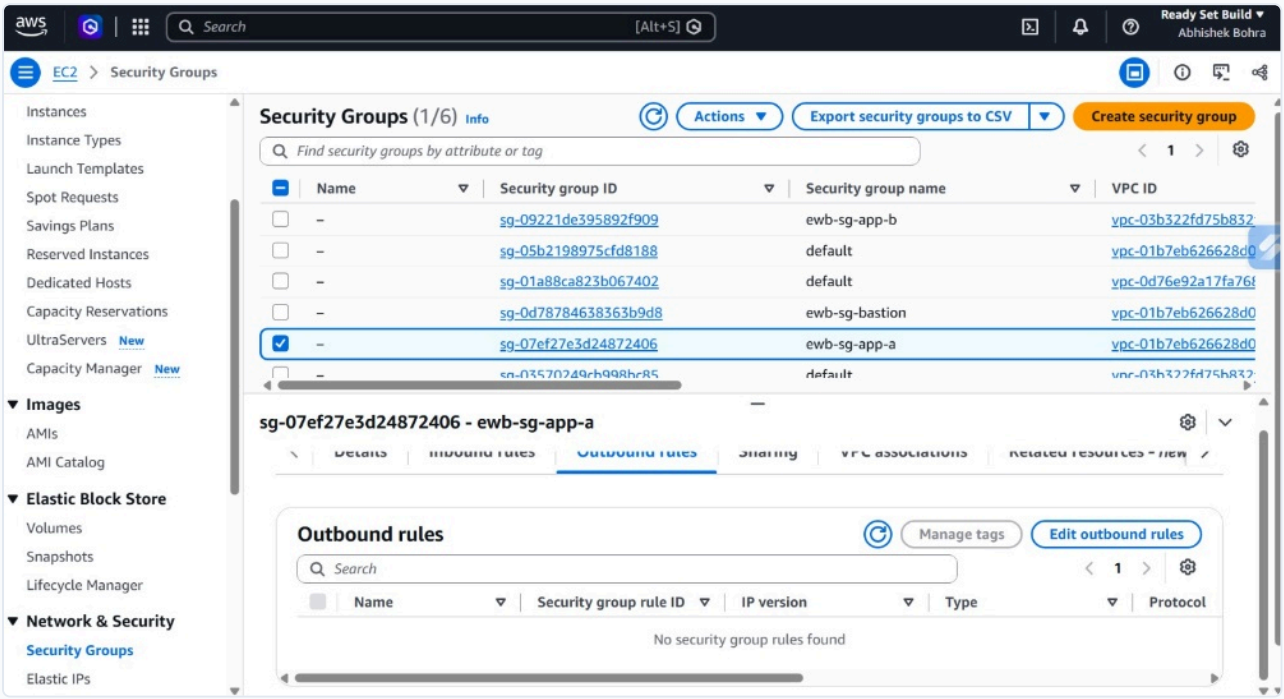
7. Troubleshooting Analysis & Controlled Failure Investigation

7.1 Real-World Root Cause Analysis: Missing Outbound Rule on ewb-sg-app-a

During initial validation, a critical behavioral property of stateful Security Groups was captured and resolved:

- **Diagnostic Observation:** While SSH ProxyJump and the reverse ping from App-B → App-A succeeded, the forward ping from App-A → App-B initially experienced 100% loss despite valid route table targets.
- **Inspection:** As verified in console evidence, `ewb-sg-app-a` (ID: `sg-07ef27e3d24872406`) was missing its default outbound rule (`All traffic, 0.0.0.0/0`).

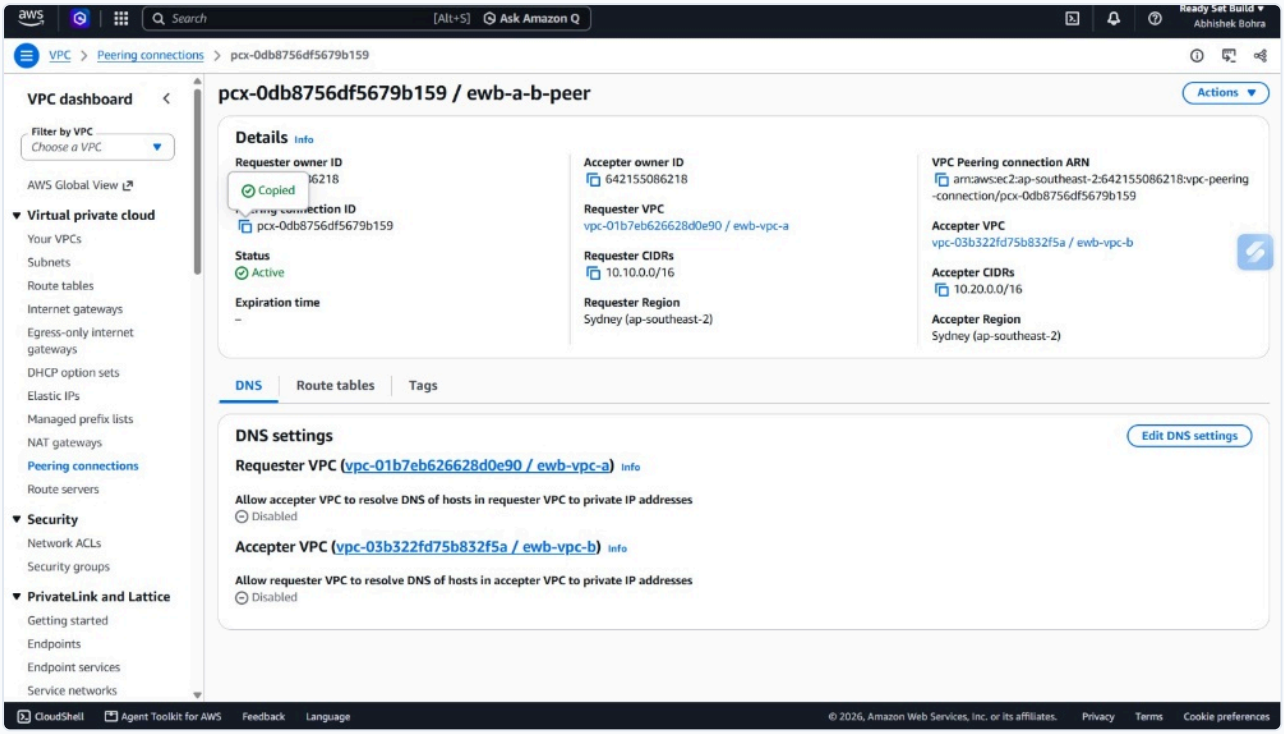
- **Networking Mechanism:** Because AWS Security Groups are stateful, return packets for incoming requests are allowed automatically. Thus, when App-B initiated the ping, App-A's reply was auto-permitted. However, when App-A attempted to *originate* an outbound ICMP request, the missing egress rule dropped the traffic at the hypervisor boundary.
- **Remediation:** Adding back the standard outbound allow-all rule restored full forward ICMP reachability immediately.



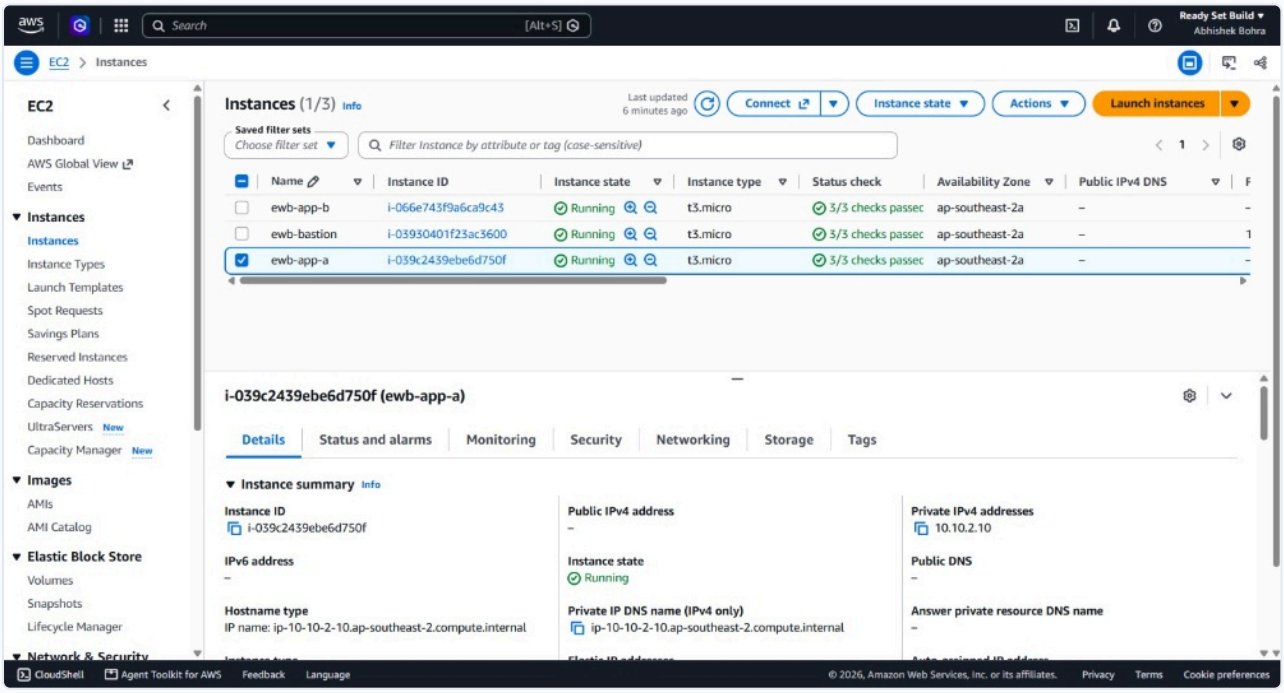
Evidence Artifact 12: Security group sg-07ef27e3d24872406 exhibiting empty outbound rules prior to diagnostic remediation.

7.2 Controlled Failure: Temporary Inbound ICMP Removal on App-B

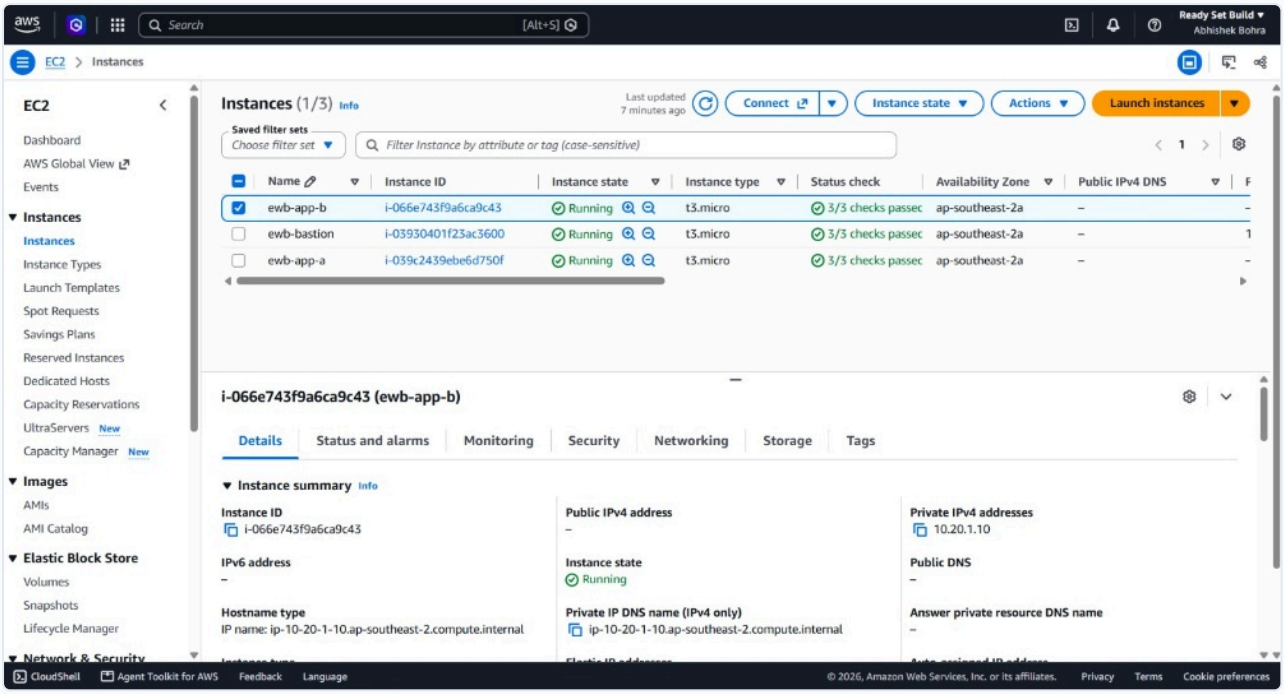
As mandated by the laboratory workbook, a controlled negative test was executed by editing `ewb-sg-app-b` (ID: `sg-09221de395892f909`) to delete the inbound ICMP Echo Request rule from `10.10.2.10/32`.



Evidence Artifact 01: Peering Connection ewb-a-b-peer (pcx-0db8756df5679b159) in Active status between vpc-01b7eb626628d0e90 and vpc-03b322fd75b832f5a.



Evidence Artifact 02: EC2 Instance ewb-app-a (i-039c2439ebe6d750f) running with private IP 10.10.2.10 and no public IPv4 address.



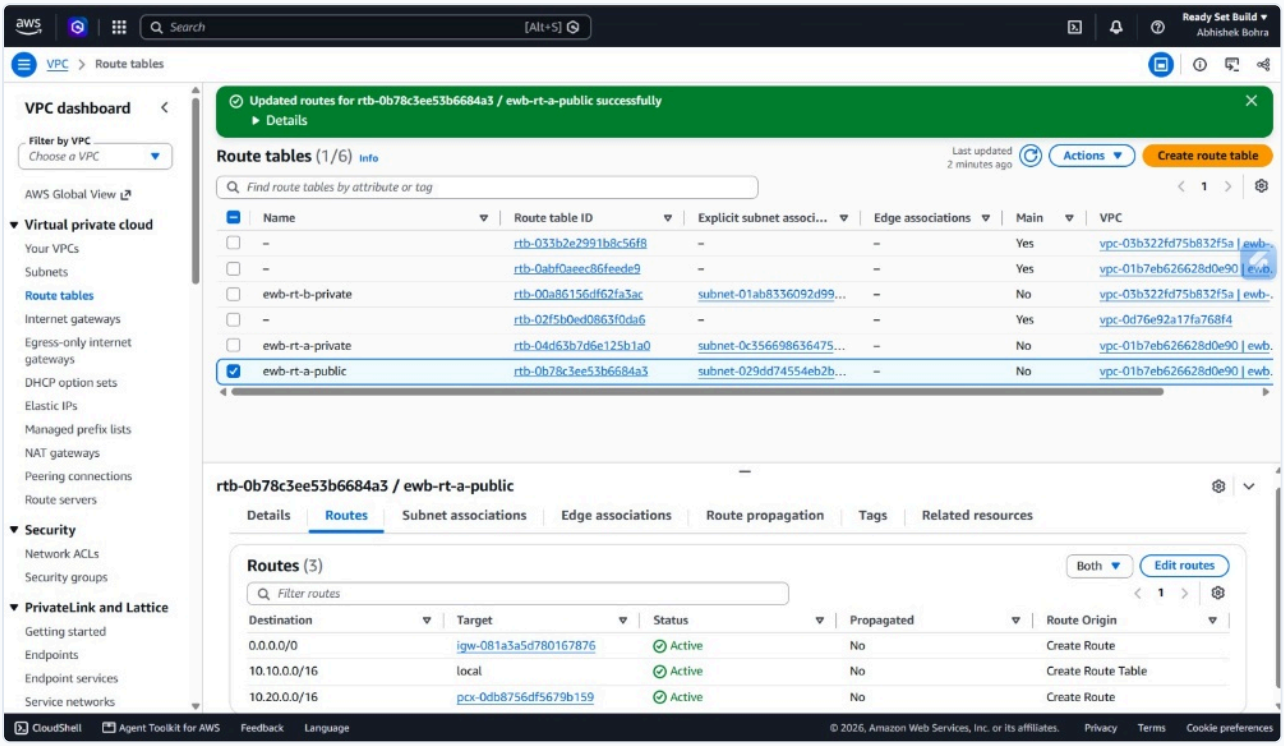
Evidence Artifact 03: EC2 Instance ewb-app-b (i-066e743f9a6ca9c43) running with private IP 10.20.1.10 and no public IPv4 address.

3. Route Tables & Subnet Association Architecture

A VPC peering connection functions solely as a passive network conduit. Active routing requires explicit entries in the respective route tables. In symmetrical routing, each VPC requires a return route back to the sender's CIDR block.

3.1 Public Route Table: ewb-rt-a-public (ID: rtb-0b78c3ee53b6684a3 | Subnet: ewb-a-public)

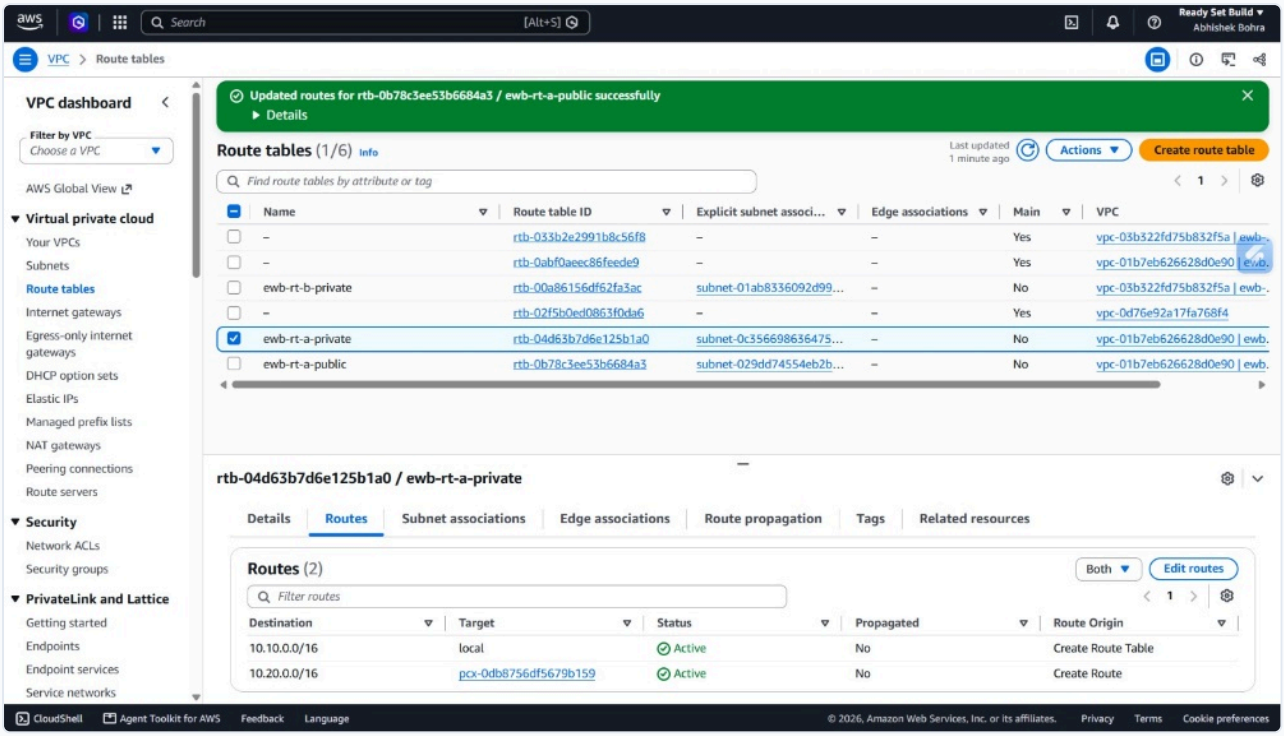
Destination CIDR	Target	Status	Architectural Purpose
10.10.0.0/16	local	Active	Intra-VPC A communication (Bastion to App-A).
0.0.0.0/0	igw-081a3a5d780167876	Active	Inbound administrator SSH & outbound internet access.
10.20.0.0/16	pcx-0db8756df5679b159	Active	Enables Bastion to establish SSH ProxyJump connection to App-B.



Evidence Artifact 04: Route table ewb-rt-a-public showing routes to local, IGW, and the peering target pcx-0db8756df5679b159.

3.2 Private Route Table: ewb-rt-a-private (ID: rtb-04d63b7d6e125b1a0 | Subnet: ewb-a-private)

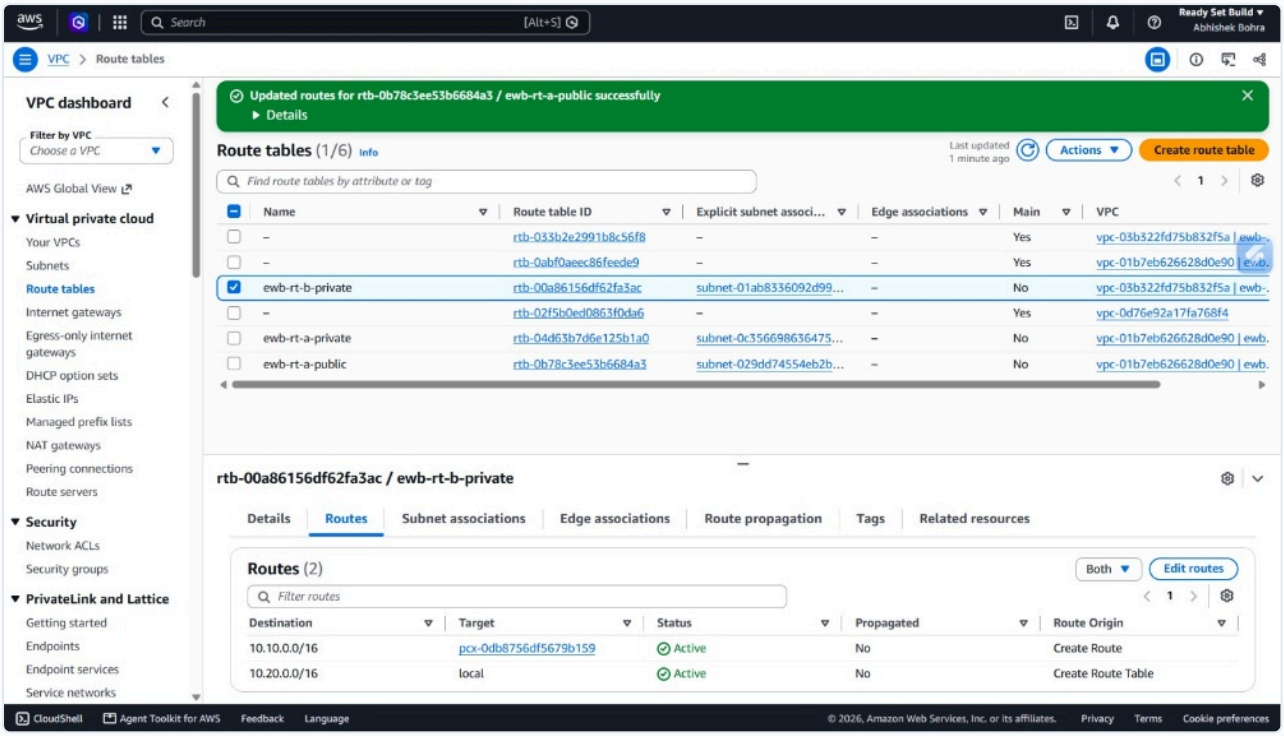
Destination CIDR	Target	Status	Architectural Purpose
10.10.0.0/16	local	Active	Local VPC communication. No internet route configured.
10.20.0.0/16	pcx-0db8756df5679b159	Active	Routes forward private traffic from App-A to VPC B workloads.



Evidence Artifact 05: Route table ewb-rt-a-private (rtb-04d63b7d6e125b1a0) with active route to 10.20.0.0/16 targeting pcx-0db8756df5679b159.

3.3 Private Route Table: ewb-rt-b-private (ID: rtb-00a86156df62fa3ac | Subnet: ewb-b-private)

Destination CIDR	Target	Status	Architectural Purpose
10.20.0.0/16	local	Active	Local VPC B communication. No internet gateway attached to VPC B.
10.10.0.0/16	pcx-0db8756df5679b159	Active	**Return Path:** Routes responses and SSH return packets back to VPC A.

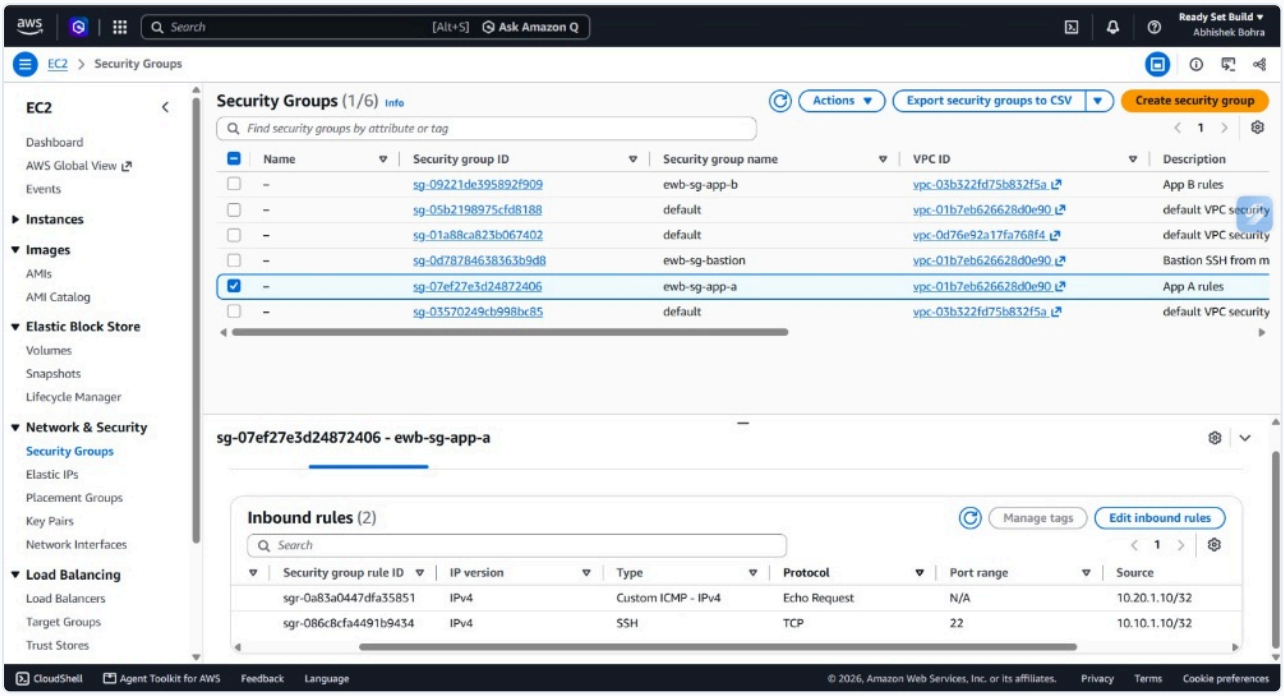


Evidence Artifact 06: Route table ewb-rt-b-private (rtb-00a86156df62fa3ac) with explicit return route to 10.10.0.0/16 targeting pcx-0db8756df5679b159.

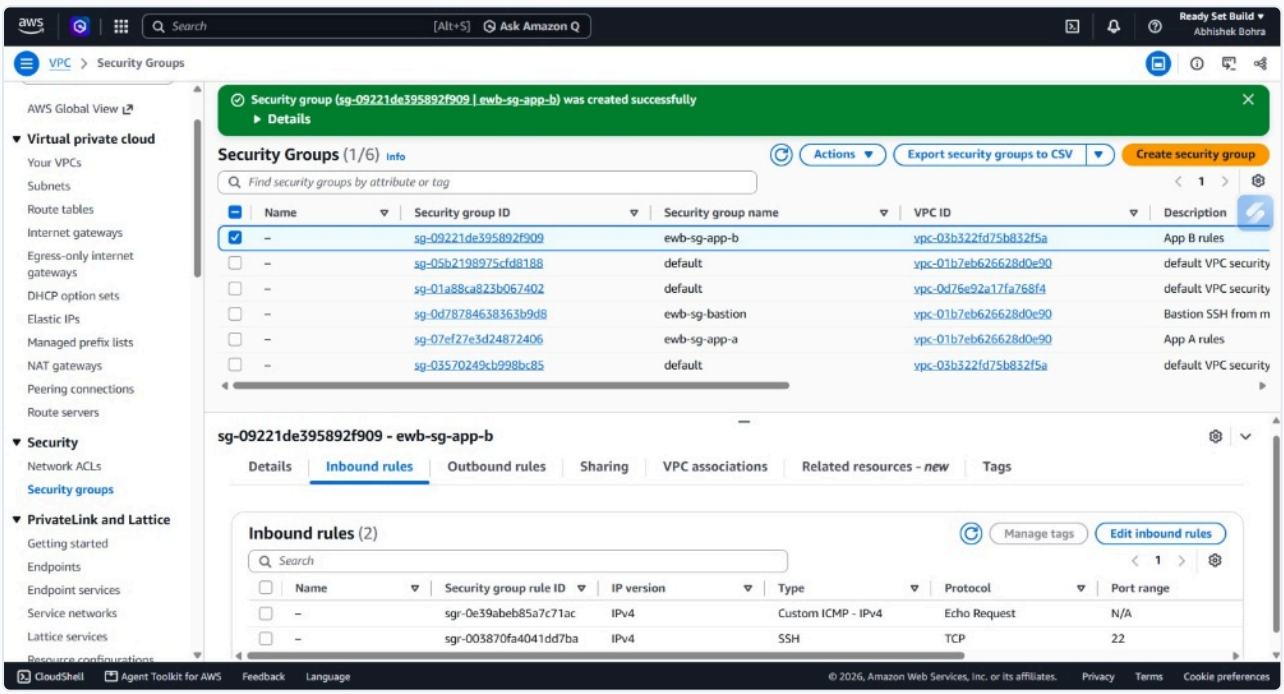
4. Security Group Configurations

In accordance with least-privilege principles, all security groups scope inbound access to specific `/32` host IPs. Outbound rules retain default `0.0.0.0/0` allow-all since AWS Security Groups are stateful.

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ewb-sg-app-a	sg-07ef27e3d24872406	Custom ICMP - IPv4	Echo Request (8/0)	10.20.1.10/32	Authorizes inbound diagnostic pings solely from App-B.
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ewb-sg-app-b	sg-09221de395892f909	Custom ICMP - IPv4	Echo Request (8/0)	10.10.2.10/32	Authorizes inbound diagnostic pings solely from App-A.



Evidence Artifact 07: Security Group ewb-sg-app-a (sg-07ef27e3d24872406) showing SSH from 10.10.1.10/32 and ICMP from 10.20.1.10/32.



Evidence Artifact 08: Security Group ewb-sg-app-b (sg-09221de395892f909) showing SSH from 10.10.1.10/32 and ICMP from 10.10.2.10/32.

5. Administration via OpenSSH ProxyJump

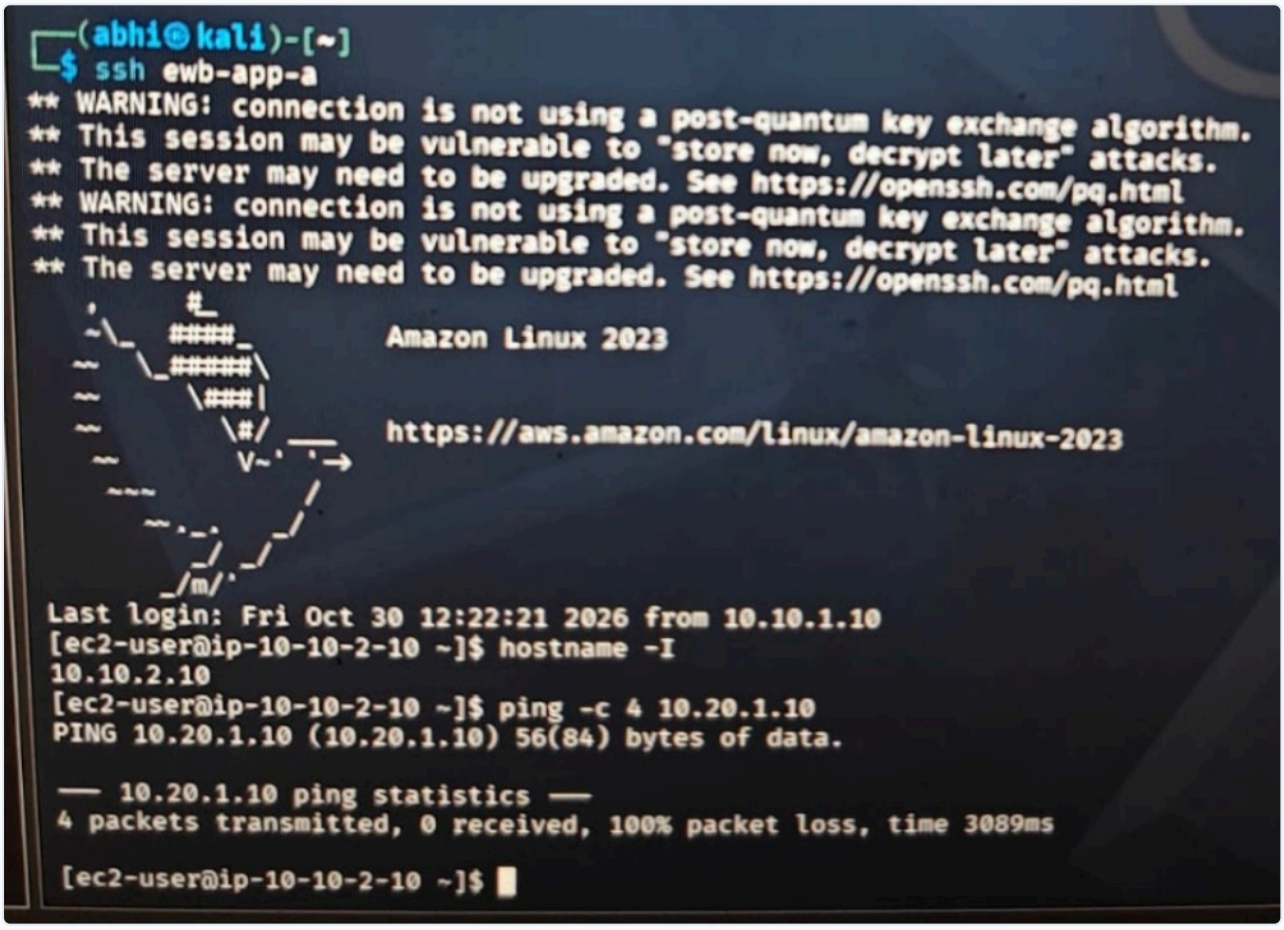
To access the completely private instances without exposing private keys on intermediate servers, OpenSSH client-side `ProxyJump` was configured on the local administrator workstation.

```
# ~/.ssh/config configuration Host ewb-bastion HostName 15.135.243.160 User ec2-user IdentityFile ~/.ssh/ewb-lab.pem IdentitiesOnly yes Host ewb-app-a HostName 10.10.2.10 User ec2-user IdentityFile ~/.ssh/ewb-lab.pem IdentitiesOnly yes ProxyJump ewb-bastion Host ewb-app-b HostName 10.20.1.10 User ec2-user IdentityFile ~/.ssh/ewb-lab.pem IdentitiesOnly yes ProxyJump ewb-bastion
```

6. Positive, Negative & Controlled Failure Verification

Test 01: Baseline Negative Test (Before Route & Peering Activation)

Executed from App-A attempting to reach App-B (`10.20.1.10`). Confirms complete network isolation prior to peering and route table entries.



Evidence Artifact 09: Baseline test on App-A showing 100% packet loss to 10.20.1.10 prior to peering configuration.

```
┌──(abhi㉿kali)-[~] └─$ ssh ewb-app-a Last login: Fri Oct 30 12:22:21 2026 from 10.10.1.10 [ec2-user@ip-10-10-2-10 ~]$ hostname -I 10.10.2.10 [ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10 PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data. --- 10.20.1.10 ping statistics --- 4 packets transmitted, 0 received, 100% packet loss, time 3089ms
```

Test 02: Forward Path Verification (App-A → App-B)

Executed from App-A after peering activation, route table population, and security group verification.

```
┌──(abhi㉿kali)-[~]
└─$ ssh ewb-app-a
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to 'store now, decrypt later' attacks.
** The server may need to be upgraded. See https://openssh.com/pg.html
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to 'store now, decrypt later' attacks.
** The server may need to be upgraded. See https://openssh.com/pg.html
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V-/.'-\
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Fri Oct 30 12:22:21 2026 from 10.10.1.10
[ec2-user@ip-10-10-2-10 ~]$ hostname -I
10.10.2.10
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
64 bytes from 10.20.1.10: icmp_seq=1 ttl=63 time=3091 ms
64 bytes from 10.20.1.10: icmp_seq=2 ttl=63 time=3088 ms
64 bytes from 10.20.1.10: icmp_seq=3 ttl=63 time=3087 ms
64 bytes from 10.20.1.10: icmp_seq=4 ttl=63 time=3089 ms

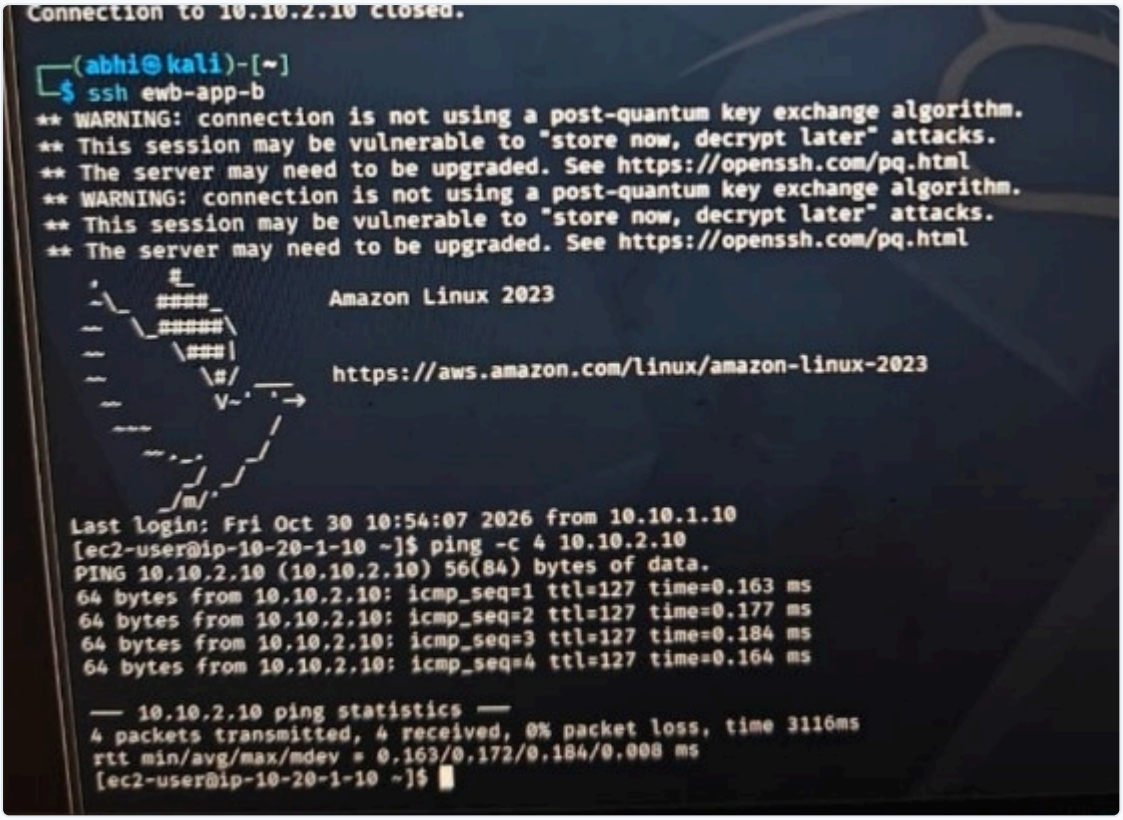
--- 10.20.1.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3089ms
[ec2-user@ip-10-10-2-10 ~]$
```

Evidence Artifact 10: Successful forward ping from App-A (10.10.2.10) to App-B (10.20.1.10) confirming 0% packet loss.

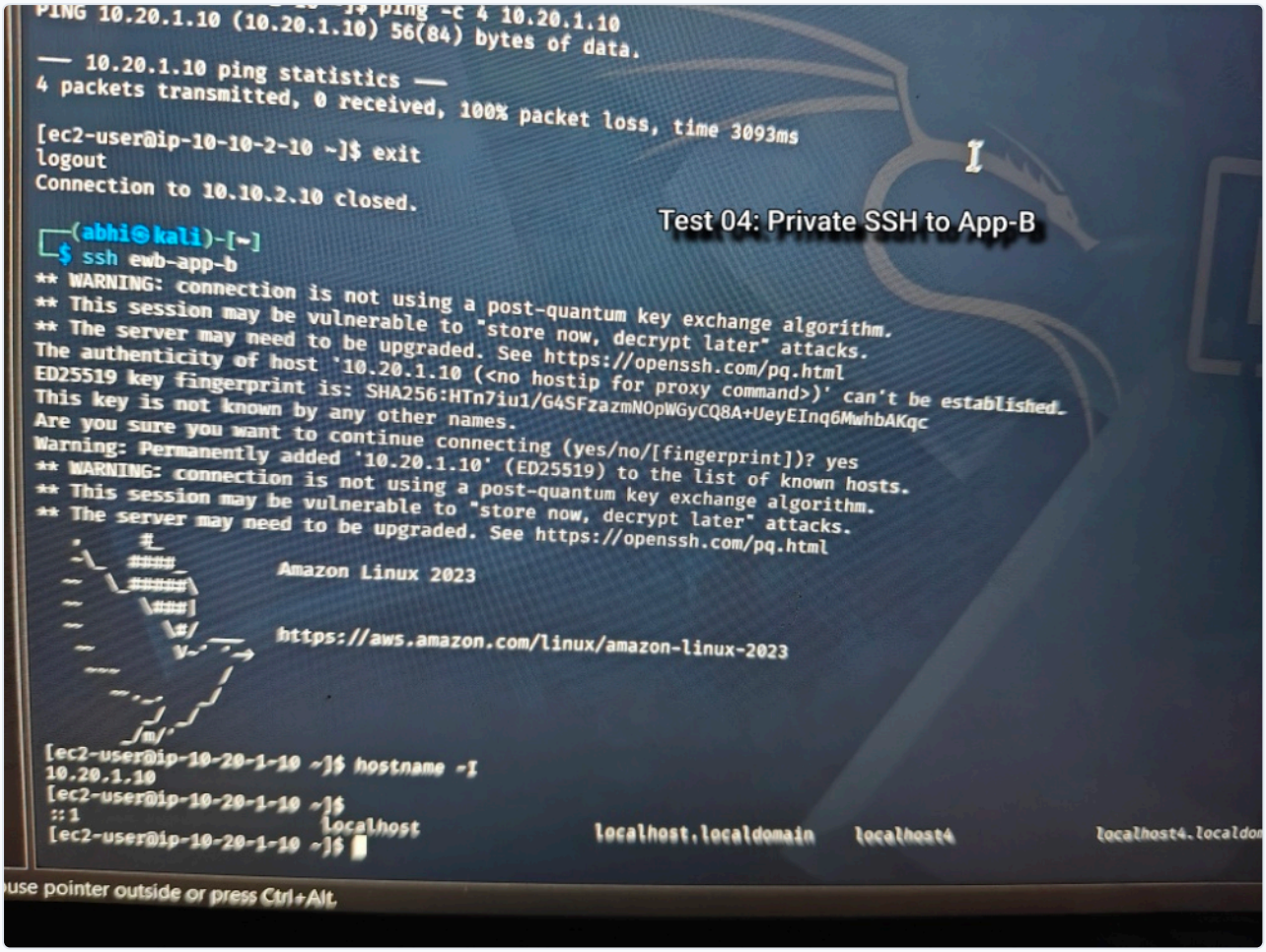

```
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10 PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data. 64 bytes from 10.20.1.10: icmp_seq=1 ttl=63 time=3091 ms 64 bytes from 10.20.1.10: icmp_seq=2 ttl=63 time=3088 ms 64 bytes from 10.20.1.10: icmp_seq=3 ttl=63 time=3087 ms 64 bytes from 10.20.1.10: icmp_seq=4 ttl=63 time=3089 ms --- 10.20.1.10 ping statistics --- 4 packets transmitted, 4 received, 0% packet loss, time 3089ms
```


Test 03 & 04: Reverse Path (App-B → App-A) & Transparent SSH ProxyJump

Executed from Kali workstation jumping directly into App-B (`10.20.1.10`) via Bastion (`10.10.1.10`), then initiating an ICMP Echo Request targeting App-A (`10.10.2.10`).



Evidence Artifact 11: Reverse ping from App-B (10.20.1.10) to App-A (10.10.2.10) proving symmetrical return connectivity and OpenSSH ProxyJump reachability.



Evidence Artifact 12: Terminal output proving direct SSH access into isolated host App-B (10.20.1.10) using ProxyJump.

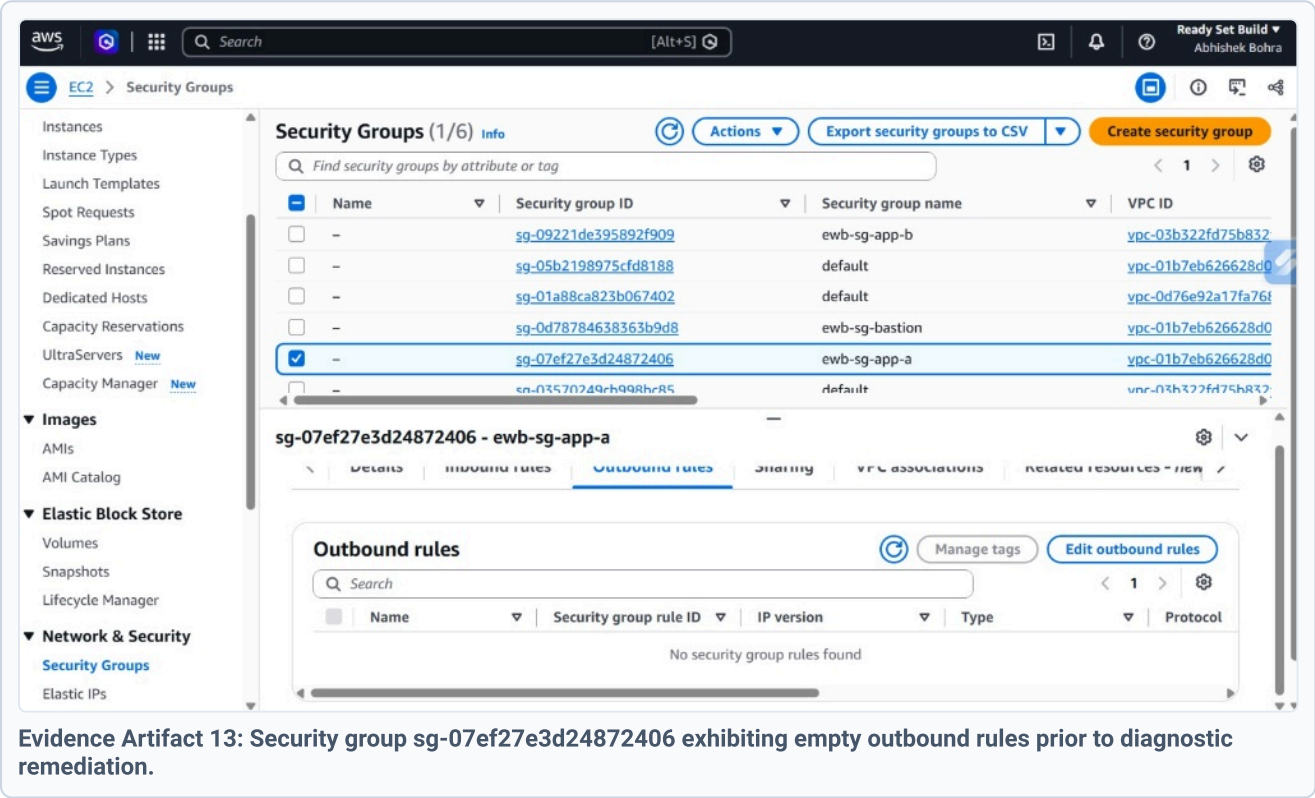
```
┌──(abhi㉿kali)-[~] └─$ ssh ewb-app-b Last login: Fri Oct 30 10:54:07 2026 from
10.10.1.10 [ec2-user@ip-10-20-1-10 ~]$ ping -c 4 10.10.2.10 PING 10.10.2.10 (10.10.2.10)
56(84) bytes of data. 64 bytes from 10.10.2.10: icmp_seq=1 ttl=127 time=0.163 ms 64 bytes
from 10.10.2.10: icmp_seq=2 ttl=127 time=0.177 ms 64 bytes from 10.10.2.10: icmp_seq=3
ttl=127 time=0.184 ms 64 bytes from 10.10.2.10: icmp_seq=4 ttl=127 time=0.164 ms ---
10.10.2.10 ping statistics --- 4 packets transmitted, 4 received, 0% packet loss, time
3116ms rtt min/avg/max/mdev = 0.163/0.172/0.184/0.008 ms
```

7. Troubleshooting Analysis & Controlled Failure Investigation

7.1 Real-World Root Cause Analysis: Missing Outbound Rule on ewb-sg-app-a

During initial validation, a critical behavioral property of stateful Security Groups was captured and resolved:

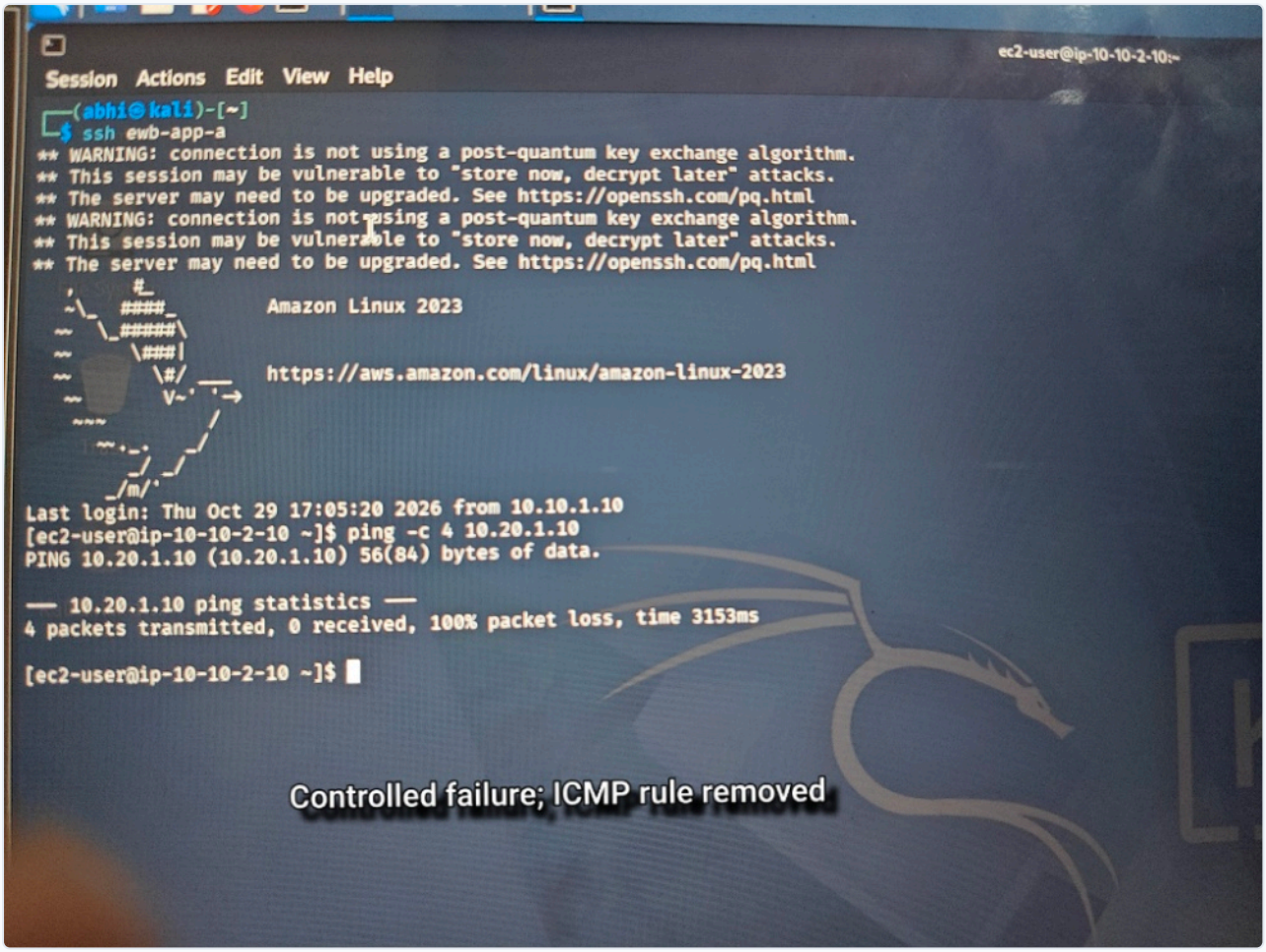
- **Diagnostic Observation:** While SSH ProxyJump and the reverse ping from App-B → App-A succeeded, the forward ping from App-A → App-B initially experienced 100% loss despite valid route table targets.
- **Inspection:** As verified in console evidence, ewb-sg-app-a (ID: sg-07ef27e3d24872406) was missing its default outbound rule (All traffic, 0.0.0.0/0).
- **Networking Mechanism:** Because AWS Security Groups are stateful, return packets for incoming requests are allowed automatically. Thus, when App-B initiated the ping, App-A's reply was auto-permitted. However, when App-A attempted to *originate* an outbound ICMP request, the missing egress rule dropped the traffic at the hypervisor boundary.
- **Remediation:** Adding back the standard outbound allow-all rule restored full forward ICMP reachability immediately.



Evidence Artifact 13: Security group sg-07ef27e3d24872406 exhibiting empty outbound rules prior to diagnostic remediation.

7.2 Controlled Failure: Temporary Inbound ICMP Removal on App-B

As mandated by the laboratory workbook, a controlled negative test was executed by editing ewb-sg-app-b (ID: sg-09221de395892f909) to delete the inbound ICMP Echo Request rule from 10.10.2.10/32.



Evidence Artifact 14: Terminal capture during controlled failure with ICMP Echo rule removed on App-B resulting in 100% packet loss from App-A.

With the rule removed, `ping -c 4 10.20.1.10` from App-A immediately stalled with 100% packet loss while administrative SSH sessions via the Bastion remained unaffected. Re-adding the rule restored connectivity with 0% loss, proving granular security isolation.

8. Technical Defense & Viva Voce

1. Why is the return route needed?

VPC routing is stateless at the subnet boundary. While a peering connection provides the link, each VPC evaluates routing independently. When App-A initiates communication to App-B, App-B generates response packets destined for VPC A's CIDR. If VPC B's route table lacks a route targeting the peering connection for 10.10.0.0/16, the packets are discarded.

2. Why is the App-B SSH source the bastion private IP?

When using SSH ProxyJump, the administrative workstation establishes an initial SSH session to the bastion. The bastion then initiates a new, separate outbound TCP connection from its private interface (10.10.1.10) to App-B's private IP (10.20.1.10). Because the TCP connection terminates at and re-originates from the bastion, App-B receives packets with the source IP 10.10.1.10.

3. Why does Active peering not guarantee a successful ping?

"Active" simply denotes that the peering agreement handshake was completed between the VPC endpoints. Packets will still be dropped if: (a) explicit subnet route table entries do not direct traffic to the `pcx` ID, or (b) Security Groups or Network ACLs filter the protocol/port.

4. Can VPC B use VPC A's Internet Gateway over the peering connection?

No. VPC Peering is strictly non-transitive and does not support edge routing to attached Internet Gateways, NAT Gateways, or Virtual Private Gateways. To route VPC B internet traffic through VPC A, a Network Firewall or transit architecture (such as AWS Transit Gateway) would be required.

9. Teardown & Account Cleanup Evidence

To prevent accidental AWS billing, all created resources were decommissioned systematically in reverse order of creation:

1. Terminated EC2 instances (`ewb-bastion: i-03930401f23ac3600`, `ewb-app-a: i-039c2439ebe6d750f`, `ewb-app-b: i-066e743f9a6ca9c43`) and confirmed attached EBS volume deletion.
2. Deleted Peering Connection (`pcx-0db8756df5679b159`) and removed all custom routes.
3. Deleted security groups (`ewb-sg-app-a`, `ewb-sg-app-b`, `ewb-sg-bastion`).
4. Detached and deleted Internet Gateway (`igw-081a3a5d780167876`) and deleted custom route tables.
5. Deleted subnets and terminated both Virtual Private Clouds (`vpc-01b7eb626628d0e90`, `vpc-03b322fd75b832f5a`).
6. Verified AWS Resource Groups and Cost Explorer: zero active billable resources in `ap-southeast-2`.